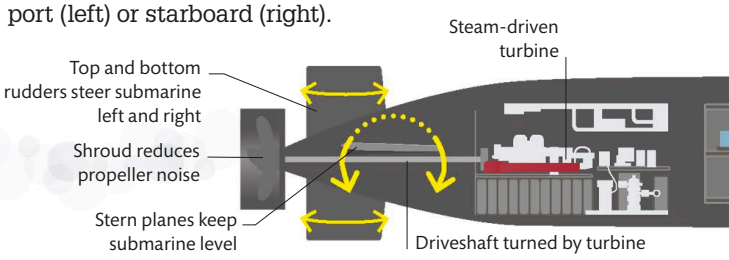
Submarines

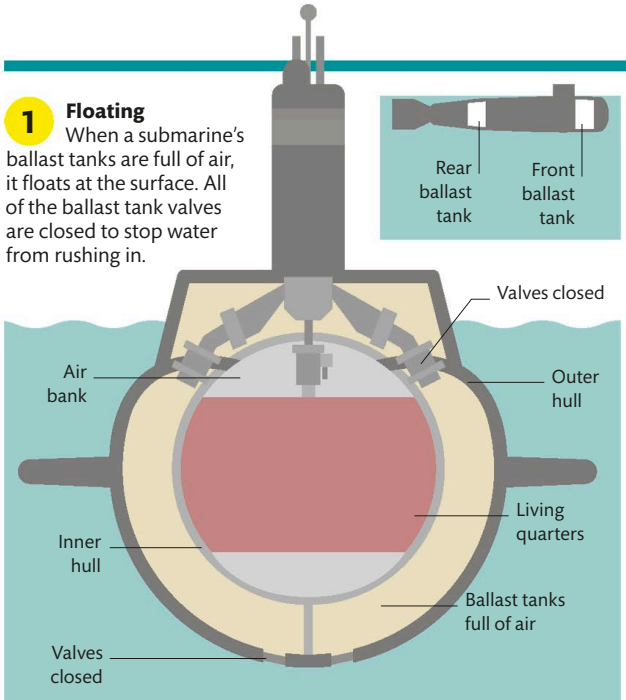
A submarine is a vessel designed to be used underwater, often for military use. Ballast tanks allow submarines to sink or float. Usually powered by a nuclear reactor or a diesel-electric engine, submarines contain high-tech navigation and communication systems and can remain hidden for months at a time.

Moving through water

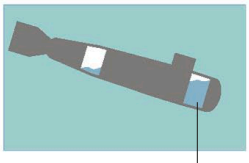
As a submarine moves through the ocean, propelled by its powerful engine, the crew steer it by moving three types of control surfaces—bow planes, stern planes, and rudders. They tilt the bow planes to make the submarine climb higher or dive deeper in the water. They adjust the stern planes to keep the submarine level. They use rudders to steer the submarine to port (left) or starboard (right).

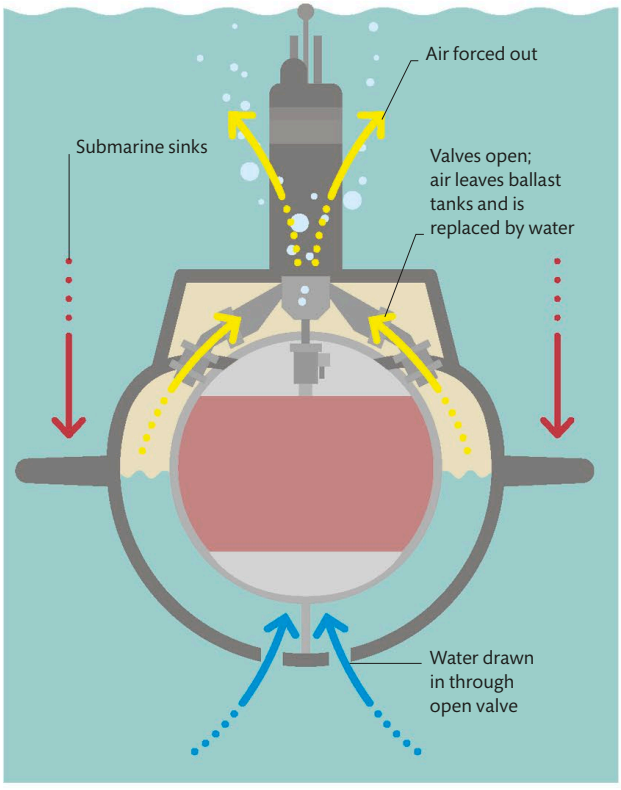


1 Floating
When a submarine's ballast tanks are full of air, it floats at the surface. All of the ballast tank valves are closed to stop water from rushing in.



2 Diving
A submarine dives by opening its ballast tank valves to let seawater flood inside. The submarine, now heavier than the same volume of water, sinks. Taking in more water makes it sink to a greater depth.





How a submarine dives and rises

Submarines are able to dive to great depths and return to the surface again because they can change their density relative to the water around them. If a submarine's density is greater than the surrounding water, it sinks. Reducing the submarine's density makes it more buoyant, so it floats up toward the surface. The crew changes the submarine's density by filling its ballast tanks, located between the inner and outer hulls, with seawater or compressed air.